W1-2-60-1-6

JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

UNIVERSITY EXAMINATION 2018/2019

SMA 2101: CALCULUS I

DATE: JULY 2019 **TIME: 2HOURS**

INSTRUCTIONS ANSWER QUESTION ONE AND ANY OTHER TWO

QUESTION ONE (30MARKS)

a) Determine the values of b and c so that the following function is continuous on the entire real line $f(x)=\begin{cases}1 & if \quad x\le 3\\ ax+b & if\ 3<x<5\\ 7 & if \quad 5\le x\end{cases}$ (5marks)

b) Evaluate the following limits

i) $\lim\limits_{x\to 1}\left(\dfrac{\sqrt{2x+1}-\sqrt{3}}{x-1}\right)$ (4 marks)

ii) $\lim\limits_{x\to 2}\left(\dfrac{1}{x-2}-\dfrac{x-2}{x^2-3x+2}\right)$ (4 marks)

iii) $\lim\limits_{x\to\infty}\left(\dfrac{3x^2-6x}{x^3}\right)$ (3marks)

c) Find the derivative by the first principle of $y(x)=\dfrac{7-3x}{20-5x}$ (6 marks)

d) Find $\dfrac{dy}{dx}$ given

i) $x^3-3x^2y^3=10\left(\dfrac{x}{y}\right)$ at $x=-2$, $y=1$ (4 marks)

ii) $y=x^4\sin^{-1}(2x)$ (4 marks)

QUESTION TWO (20 MARKS)

a) The parametric equation of a curve are given by $x = \cos^3 t$ and $y = \sin^3 t$ .Find the equation of the normal line at the point where $t = \frac{\pi}{4}$ (6 marks)

b) Find $\frac{dy}{dx}$ given

(i) $y = \frac{x}{\ln(x+3)} - xe^{x^2}$ (4 marks)

(ii) $y = 2^x \tan^{-1}(2x)$ (4 marks)

c) The revenue for the number of software suits produced and sold by Dell Company is given by $R = 90x - x^2$ where x is the number of units produced and sold daily. The cost of producing 1 unit is 50 Euros. What is the most profitable number of produce? (6 marks)

QUESTION THREE (20 MARKS)

(a) If $x = \frac{2-3t}{1+t}$ and $y = \frac{3+2t}{1+t}$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at x=2 (8 marks)

(b) Given that $x^2 - xy + y^2 = 3$. Show that $\frac{dy}{dx} = \frac{2x-y}{x-2y}$ and $\frac{d^2y}{dx^2} = \frac{18}{(x-2y)^3}$ (7 marks)

(c) Under a perfect competition the price as sh 6 per unit has been determined. An individual firm has a total cost function given by $C = 10 + 15x - 5x^2 + \frac{x^3}{3}$.To find the quantity produced at which profit will be maximum profit. (5marks)

QUESTION FOUR (20 MARKS)

(a) Find $\frac{dy}{dx}$ from the first principle for $y = \tan x$ (7 marks)

(b) Sketch the graph of the curve whose equation is $y = \frac{x-2}{x-1}$ (7 marks)

(c) Determine the derivative of $y = \ln\left[\frac{e^{-x^2}}{\sqrt{x^2-1}}\right]$ (6 marks)